The maximum number of maximal cliques in uniform hypergraphs

Organized by Xiongfeng Zhan

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Abstract

We disprove a conjecture of Tomescu (1981) on the maximum number of maximal cliques possible in a d-uniform hypergraph. The counterexample was found by DeepSeek-V4-Pro itself without any prompting.

Keywords: hypergraph, maximal clique

1 Introduction

In 1965, Moon and Moser [3] proved that $f(n) = \max_{n=n_1+\ldots+n_k} \prod_i n_i$ is the maximum number of maximal cliques possible in a graph with n vertices. It is natural to find the analogous function $f(n,d)$ for d-uniform hypergraphs on n vertices. In 1981, Tomescu [4] gave the complete chromatic hypergraph construction and conjectured that the actual value of $f(n,d)$ is achieved by it.

Conjecture 1.1 (Tomescu [4])**.** *Let $f(n,d)$ be the maximum number of maximal cliques possible in a d-uniform hypergraphs on n vertices. Then $f(n,d) = \psi(n,d)$, where*

$$\psi(n,d) := \max_{n=n_1+\ldots+n_k} \prod_i \binom{n_i}{d-1}.$$

Moon and Moser's theorem is exactly the conjecture for $d = 2$. One can see [2, 1] for some attempts on Conjecture 1.1. In this paper, we disprove Tomescu's conjecture.

2 A counterexample for $d = 3$

A *d-uniform hypergraph* H is a pair $(V(H), E(H))$, where $V(H)$ is the *vertex set* and $E(H)$ is a collection of d-subsets of $V(H)$, called *hyperedges*. When the vertex set is clear from context, we also refer to the hypergraph simply by its hyperedge set. A *clique* in H is a subset $C \subseteq V(H)$ such that every d-subset of C is a hyperedge. Note that every subset of $V(H)$ with size less than d is trivially a clique. A clique $C \subseteq V(H)$ is *maximal* if there is no vertex $v \in V(H) \setminus C$ such that $C \cup \{v\}$ is also a clique. Denote by $c(H)$ the number of maximal cliques in the hypergraph H.

Email address: zhanxfmath@163.com (X. Zhan).

Theorem 2.1. *Conjecture 1.1 is false.*

Proof. In the following example, one can check that every 4-subset of $[7]$ contains at least one element in $F(3,7)$. Therefore, the complement hypergraph $H(3,7) = \binom{[7]}{3} \setminus F(3,7)$ has no 4-clique and then has at least $\binom{7}{3} - |F(3,7)| = 23$ maximal cliques, which is greater than $\psi(7,3) = 21$.

Example. $F(3,7)$ is the following 12 triples on $[7]$:

$$\{123, 145, 246, 127, 156, 346, 137, 467, 235, 237, 257, 357\}.$$

□

Remark 2.2. *For each $4 \le d \le 8$, a greedy algorithm is used to construct a hitting set yielding a counterexample to Conjecture 1.1. The five sets are provided in the file* `tomescu_hitting_sets.json`*, which are verified by the program* `verify_tomescu_counterexamples.py`*. In fact, DeepSeek-V4-Pro also claims that there is a counterexample for every $d \ge 9$, but its proof is incorrect. Another AI model provides a correct proof, see the Appendix.*

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References

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A Appendix

In this section, we show Conjecture 1.1 is false for every $d \ge 6$.

Lemma A.1. *Let $L(n,d) = \frac{d}{d+1}(n-d)^{-1/d}\binom{n}{d}$. Then $f(n,d) \ge L(n,d)$.*

Proof. Let G be the d-uniform random hypergraph on vertex set $[n]$, where each d-set of $[n]$ is a hyperedge independently with probability $p = 1-q$. For every $(d+1)$-set which containing no hyperedge of G, add one of its d-sets to $E(G)$. This produces a d-uniform random hypergraph F satisfying that every $(d+1)$-set of $[n]$ contains a hyperedge of F. The expected size of F is

$$\mathbb{E}(|F|) \le (1-q)\binom{n}{d} + q^{d+1}\binom{n}{d+1}.$$

So the complement $H = \binom{[n]}{d} \setminus F$ has no $(d+1)$-clique, hence every hyperedge of H is a maximal clique. Therefore

$$\mathbb{E}(c(H)) \geq \mathbb{E}(|H|) \geq q\binom{n}{d} - q^{d+1}\binom{n}{d+1}.$$

Let $q = (n-d)^{-1/d}$, we get $\mathbb{E}(c(H)) \geq L(n,d)$. So there must exist some d-uniform hypergraph H_0 such that $c(H_0) \geq L(n,d)$. □

Lemma A.2. *Let $t = \lceil 2\log d \rceil$ and $m = 2d + t$. Then for $d \geq 6$, we have $\psi(m,d) = \binom{m}{d-1}$.*

Proof. The cases $d \in \{6,7,8\}$ are straightforward. We now assume $d \geq 9$. Then $m < 3(d-1)$. Hence any partition of m has at most two parts of size at least $d-1$. If two such parts occur, we write them as $d-1+r$ and $d-1+s$, where $r, s \geq 0$ and $r+s < d-1$. We have

$$\begin{aligned}
\binom{d-1+r}{d-1}\binom{d-1+s}{d-1} &= \binom{d-1+r}{r}\binom{d-1+s}{s} \\
&\leq \binom{2(d-1)+r+s}{r+s} \\
&\leq \binom{2(d-1)+r+s}{d-1} \\
&= \binom{m}{d-1},
\end{aligned}$$

where the last inequality uses that binomial coefficients increase up to the middle. Hence, $\psi(m,d) = \binom{m}{d-1}$ for $d \geq 6$. □

Theorem A.3. *For every $d \geq 6$, Conjecture 1.1 is false.*

Proof. Using Lemma A.1, it suffices to compare $L(m,d)$ with $\psi(m,d)$. The cases $d \in \{6,7,8\}$ are straightforward. We now assume $d \geq 9$. By Lemma A.2,

$$\begin{aligned}
\log\left(\frac{L(m,d)}{\psi(m,d)}\right) &= \log\left(\frac{m-d+1}{d+1}(m-d)^{-1/d}\right) \\
&= \log\left(\frac{d+t+1}{d+1}\right) - \frac{1}{d}\log(d+t) \\
&\geq \log\left(1 + \frac{2\log d}{d+1}\right) - \frac{1}{d}\log(d + 2\log d + 1).
\end{aligned}$$

Set $T = \log d$. Using the elementary bounds $x \geq \log(1+x) \geq x - \frac{x^2}{2}$ for $x > 0$, we have

$$\begin{aligned}
\log\left(\frac{L(m,d)}{\psi(m,d)}\right) \geq & \frac{2T}{d+1} - \frac{2T^2}{(d+1)^2} - \frac{1}{d}\left(T + \log(1 + \frac{2T+1}{d})\right) \\
\geq & \frac{2T}{d+1} - \frac{2T^2}{d^2} - \frac{1}{d}\left(T + \frac{2T+1}{d}\right) \\
= & \frac{T}{d}\left(\frac{2d}{d+1} - \frac{2T}{d} - 1 - \frac{2}{d} - \frac{1}{dT}\right) \\
= & \frac{T}{d}\left(1 - \frac{2}{d+1} - \frac{2T}{d} - \frac{2}{d} - \frac{1}{dT}\right) \\
\geq & \frac{T}{d}\left(1 - \frac{2}{9+1} - \frac{2\log 9}{9} - \frac{2}{9} - \frac{1}{9\log 9}\right) \\
> & \frac{T}{d} \times 0.038 > 0,
\end{aligned}$$

where at the penultimate step we use the fact that $\frac{2}{d+1}$, $\frac{2\log d}{d}$, $\frac{2}{d}$, and $\frac{1}{d\log d}$ are all decreasing functions of d for $d \geq 9$. Therefore, $L(m,d) > \psi(m,d)$ for every $d \geq 6$. $\square$